



TANMAY SINGH

Phone: [+91-9304444091](tel:+91-9304444091) | **Email:** tanmaysingh8246@gmail.com
LinkedIn: [linkedin.com/in/tanmay-singh-28395b345](https://www.linkedin.com/in/tanmay-singh-28395b345)
GitHub: <https://github.com/TSR0705>
Location: Kattankulathur, Chennai, Tamil Nadu – 603203, India

SUMMARY

Second-year Computer Science Engineering student specializing in Cloud Computing at SRMIST, with practical experience building backend systems and CI/CD workflows using Node.js, Docker, Kubernetes, MongoDB, and Jenkins. Developed secure authentication and role-based access control (RBAC) mechanisms and worked on containerized applications and distributed systems projects. Strong interest in DevOps, backend engineering, and cloud-native development.

EDUCATION

SRM Institute of Science and Technology, Kattankulathur — Chennai	2024 - 2028
B.Tech – Computer Science & Engineering, Cloud Computing Specialization CGPA: 9.37/10	
Sarla Varma Bal Vidya Mandir, Bihar	2024
Class XII CBSE Percentage: 77.6%	
Sainik School Rewari, Haryana	2022
Class X CBSE Percentage: 91%	

INTERNSHIP EXPERIENCE

Backend Engineering Intern	Dec 2025 - Jan 2026
MedGency (Health-Tech Startup)	Chennai, Tamil Nadu

Tech: Node.js, Express.js, TypeScript, MongoDB, Redis, JWT, Jest

- Built a multi-tenant authentication and RBAC system with 4-tier role hierarchy, enforcing invite-only access and strict organization isolation.
- Developed OTP-based login and JWT authentication with Redis-backed revocation and 15-minute expiry, strengthening session security and real-time access control.
- Implemented 256-bit hashed magic-link onboarding, rate limiting, and mandatory audit logging to ensure secure, traceable, and fail-closed authorization flows.

PROJECTS

OpenCI Runner — Secure Public CI Execution Platform	Dec 2025 – Jan 2026
--	----------------------------

Project Link: [OpenCI Runner — GitHub](#)

Tech: Node.js, Express.js, EJS, Docker, Jenkins, Jest, Supertest, Git, GitHub, Linux

- Designed a secure, containerized CI execution architecture using isolated Docker runners with CPU/memory limits, network sandboxing, and automated teardown to enable safe and scalable multi-tenant code execution.
- Built a Jenkins-based CI/CD pipeline to automate testing, Docker image versioning, and release workflow orchestration following production-grade DevOps best practices.

LoadLab + DeployBot — Controlled Kubernetes Orchestration System

Nov 2025 - Dec 2025

Project Link: [LoadLab + DeployBot — GitHub](#)

Tech: Next.js (TypeScript), Kubernetes, Docker, @kubernetes/client-node, Clerk, Tailwind CSS, Bun

- Engineered a concurrency-safe Kubernetes control plane using mutex-based serialized execution and priority scheduling to eliminate race conditions and guarantee deterministic infrastructure state.
- Implemented secure Kubernetes API integration with strict namespace isolation, deployment scoping, and replica caps to enforce safe, controlled infrastructure mutations.
- Enabled real-time cluster observability, demonstrating pod scaling, readiness/liveness probes, and self-healing behavior through a chat-driven orchestration interface.

Webloom — Automated Web Monitoring Platform

Sept 2025 - Present

Project Link: [Webloom — GitHub](#)

Tech: Node.js, Express, RabbitMQ, MongoDB Atlas, Docker, Next.js, TailwindCSS, Chart.js

- Developed a distributed monitoring pipeline using RabbitMQ worker queues to continuously crawl target pages and detect content, UI, and price deviations at scheduled intervals.
- Architected versioned storage in MongoDB enabling structured snapshots, historical diffs, and timeline analysis for each monitored URL.
- Delivered a Next.js dashboard to manage jobs, inspect searchable event logs, and visualize price/change trends in real time.

POSITIONS OF RESPONSIBILITY

CADES — Member Relations & Recruitment Officer

Oct 2025 – Present

- Led recruitment and onboarding by interviewing 50+ candidates and selecting 10 members.
- Maintained structured communication flow for event and project coordination.

Founders Club SRM — Technical Member

Sep 2025 – Present

- Supported development workflows for club events such as Plan-C, IdeaSpark, and Bootcamp initiatives.
- Worked on technical implementations for idea submissions and internal automation needs.

Team SRM Hackathon — Technical Member

Aug 2025 – Present

- Helped organize Semicolon 11.0 national-level hackathon with tech support and deployment coordination.
- Troubleshoot platform issues during event execution ensuring smooth operations.

ADDITIONAL INFORMATION

Technical Skills:

- **Programming:** JavaScript, Java, C, Python
- **Cloud & DevOps:** Docker, Kubernetes, Jenkins, RabbitMQ
- **Backend:** Node.js, Express.js
- **Frontend:** Next.js, React, HTML, CSS, Tailwind CSS, Framer Motion
- **Databases:** MongoDB, MySQL
- **Tools & Platforms:** Postman, Git, GitHub, VS Code, IntelliJ IDEA

Languages:

- English, Hindi

Certifications:

- [AWS Cloud Architecting](#)
- [MongoDB Node.js Developer Path](#)
- Google Cloud Gemini Specialization Track ([Application Dev](#), [Cloud Architecture](#), [Security](#), [SDLC](#))
- [IBM Cloud Essentials](#)

Awards & Achievements:

- **Winner** – [Ossome Hacks 2.0 \(Best Freshers Team\)](#)
- **3rd Place** – [SRM Java Expo 2025](#)